

Qualifying Exam in Microeconomics
Arizona State University
August 2023

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. An exchange economy has one physical good and two states of the world. The *aggregate* endowment of the good in state s is $\omega_s > 0$ for $s = 1, 2$.

The economy has two consumers. Consumer i has subjective-expected-utility preferences on the consumption set $\mathbb{R}_+^2$ represented by

$$U_i(x^i) = \pi_1 u_i(x_1^i) + \pi_2 u_i(x_2^i)$$

where

- $x^i = (x_1^i, x_2^i) \geq 0$ and x_s^i is person i 's consumption of the good in state s ;
- $\pi_1 \in (0, 1)$ is the common probability that the state is $s = 1$ and $\pi_2 = 1 - \pi_1$; and
- $u_i : \mathbb{R}_+ \rightarrow \mathbb{R}$ is strictly concave and C^1 with $u_i' > 0$.

- (a) Define precisely the notion of a (strongly) Pareto optimal allocation for this economy.

To write what we hope is obvious, an Edgeworth box diagram might help you outline your arguments in parts (b)-(e).

- (b) Suppose that $\omega_1 = \omega_2$, that is, the aggregate endowment of the good is the same in each state. Confirm that the set of Pareto optimal allocations is

$$A^* = \{(x^1, x^2) \in \mathbb{R}_+^4 \mid x^1 = (y, y), x^2 = (\omega_1 - y, \omega_2 - y), 0 \leq y \leq \omega_1 (= \omega_2)\},$$

and *carefully* justify this conclusion.

- (c) Suppose that $\omega_1 > \omega_2$. Prove that, at any *positive* Pareto optimal allocation, $x_1^i > x_2^i$ for $i = 1, 2$. (By “positive” we mean that $x^i \gg 0$ for $i = 1, 2$.)
- (d) Succinctly describe in words the differences between the conclusions and hypotheses of parts (b) and (c).
- (e) Now remove the qualifier “positive” in part (c). Does the conclusion still hold? If so, prove it, or explain why your proof in (c) applies to this case; if not, give a counter-example, the simpler the better.

2. In the town of Lake Wobegone, there are three contiguous houses that belong to very competitive neighbors, indexed by $i \in \{A, B, C\}$. Each neighbor builds a swimming pool in her property. Neighbor i 's preferred pool length depends continuously on the pool lengths chosen by the other two neighbors. Given the lengths chosen by the neighbors, i has a single most favorite length for her pool.

Note that a neighbor may want to showcase the longest swimming pool, or she may be satisfied to have the average-length swimming pool, or any other rule expressing any preferences provided the conditions mentioned in the previous paragraph are satisfied.

Lake Wobegone limits the length of private pools to a maximum of 50 meters (an olympic pool). The minimum acceptable pool length for all three neighbors is 7 meters.

The vector of pool-lengths chosen (one length by each neighbor) is an equilibrium, when no neighbor that knows the vector wants to deviate from her chosen length.

If under the conditions described there is an equilibrium, prove it. If under the conditions describe there might not be an equilibrium, please provide a counter example.

3. Consider the following version of Spence's signaling model. A worker has one of a finite number of types, $x_1 < x_2 < \dots < x_N$. Assume that $(\forall n, 1 < n \leq N) x_n - x_{n-1} = 2$.

A worker of type x_n generates value x_n to the firm. A worker's type is not observable to the firm. The worker chooses education level $e \geq 0$; education is costly and does not change the productivity of the worker. A worker of type x_n who chooses education level e and receives wage w derives a *net* utility of $u(x_n, e, w) = w - k(x_n)e^2$ where $k(x_n) > 0$ is a decreasing strictly positive function of x_n . Wages offered satisfied the zero-profit condition, that is to say, w equals the expected value of the worker to the firm given the available information in equilibrium.

- (a) Verify the single-crossing property; geometrically this property states that the indifference curves in (e, w) -space of different types cross at most once.
- (b) Construct a separating equilibrium in which type x_n chooses $e^*(x_n)$ by identifying $e(x_1)$ and providing a recursion that specifies $[e^*(x_n)]^2$ as a function of $[e^*(x_{n-1})]^2$ for each $n > 1$.

4. The market inverse demand for a good is $P(Q) = A - Q$ for *all* $Q \geq 0$, where $Q \geq 0$ is the total production of the good. (Note that, if aggregate output is greater than A , the price is negative. As it turns out, this rather unsatisfactory assumption does not affect the conclusion, but simplifies the analysis.)

There are two firms who can produce the good. Firm 2's cost is known to be $c_2(q_1) = kq_2$. Firm 1's cost is $c_1(q_1, \theta) = (k + \theta)q_1$, where θ is firm 1's cost *type*. Each firm i 's output can be any nonnegative number.

Each firm i is risk neutral, and so only cares about its expected profit, the higher the better.

Consider this Bayesian game. First, Nature chooses firm 1's type from $\{\theta_L, \theta_H\}$, where $\theta_L < 0 < \theta_H$. The two types are equally likely and the expected value of θ is *zero* (so that the *expected* unit cost of firm 1 equals firm 2's unit cost of k). Nature reveals θ to firm 1, but not to firm 2. The firms then play a Cournot duopoly game.

Assume that

$$\frac{A - k}{2} > \theta_H (> 0 > \theta_L) \quad \text{and} \quad k + \theta_L > 0.$$

The second inequality simply ensures that a type- θ_L firm 1's unit cost is positive.

- (a) Carefully describe the strategy sets of firms 1 and 2 in this game. Define precisely the notion of a Bayesian-Nash equilibrium (BNE) for this game.
- (b) Derive firm 2's best reply correspondence for this game, and *carefully* explain your derivation.
- (c) Is there a BNE in which firm 2's output is zero? Is there a BNE in which either type of firm 1's output is zero? In each case, explain.
- (d) Identify all BNE of the game.